QMWS Spring 2026: Introduction to Python for Data Analysis

Day One: Foundations

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Section 1

Installation

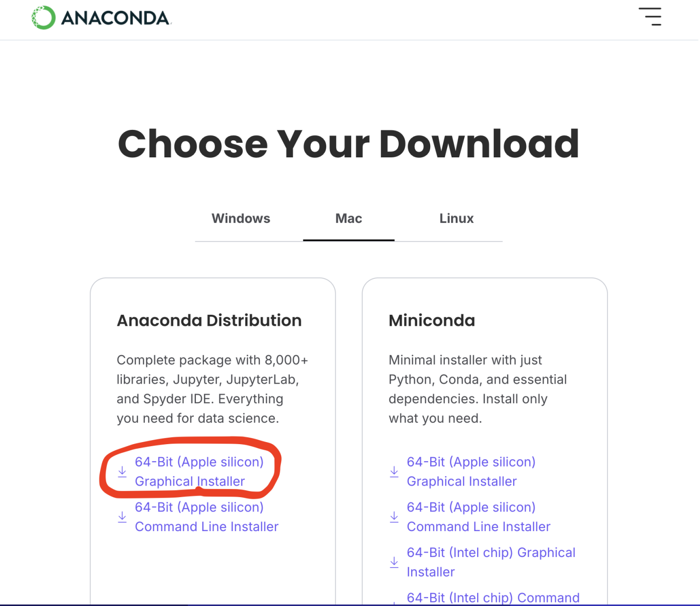
Anaconda

- A tool for configuring Python **environments**
  - More often than not, you need to install **packages** that allow you to code more easily/efficiently
  - Allows for version control
  - Issues can arise if environments are not used, for example:
    - If you install two packages that conflict with one another
    - If you install a package that requires a specific version of Python
  - Anaconda allows these issues to be isolated to a specific environment, rather than affecting your whole Python installation
    - Especially useful if you have more than one project going on

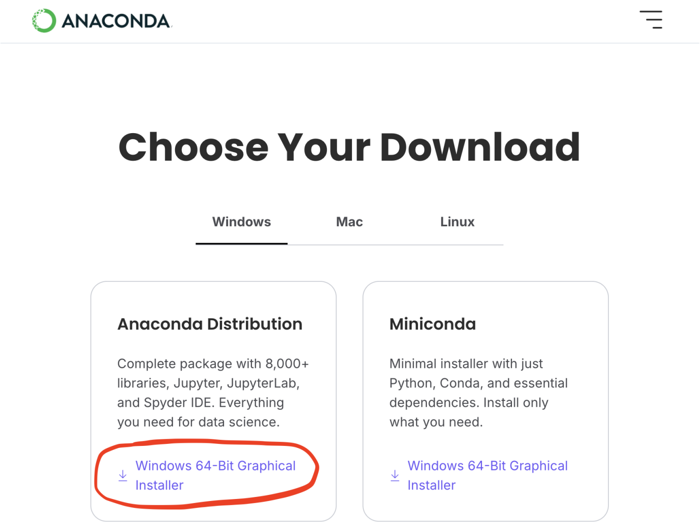
Anaconda

- Installation link: https://www.anaconda.com/download
- The simplest way to use Anaconda is via the **Navigator**
- To install:
  1. Make an account
  2. Install the full distribution via the "Graphical Installer"

- Installation link: https://www.anaconda.com/download
- On Mac:



- Installation link: https://www.anaconda.com/download
- On Windows:

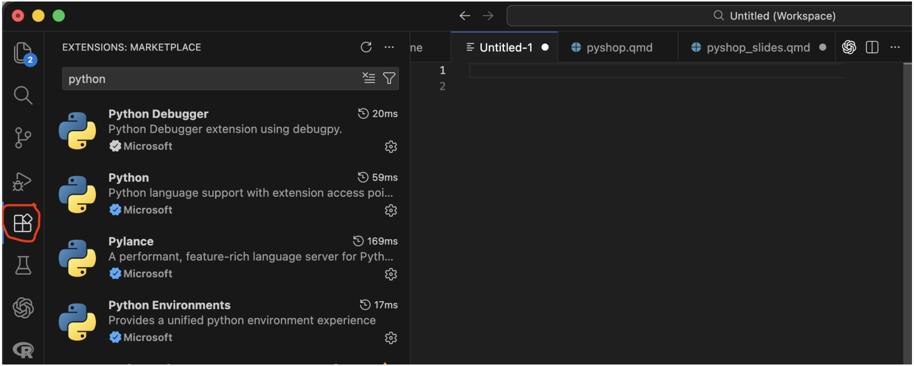


Visual Studio (VS) Code

- Installation link: https://code.visualstudio.com/download
- Integrated Development Environment (IDE)
- Can use it to code in many different languages
- Commonly used in both academia and industry

Visual Studio (VS) Code

- Once installed, go to the "Extensions" tab and search "Python"
- Install the below extensions:



Visual Studio (VS) Code

Addition after first session (June 26th, 2026):

- Make sure to install the **Jupyter** extension as well, which will allow you to use Jupyter notebooks in VS Code

Stop here!

If you:

1. Are able to open the **Anaconda Navigator**
2. Have installed **VS Code and the listed extensions**

You are ready for the workshop! We will cover next steps at the beginning of the first session.

Else, please email either of us for help:

- Gavin (gklorfin@yorku.ca)
- Bora (laze@yorku.ca)

Setting Up a Conda Environment

- See the presenter's shared screen
  - We will install the **ipykernel** package, allowing us to use **Jupyter notebooks** in VS Code (more on this in a few minutes)

Visual Studio (VS) Code

- To point VS Code to your new Conda environment:
  1. Open the Command Palette
     a. `Ctrl + Shift + P` on Windows
     b. `Command + Shift + P` on MacOS
  2. Search "Python: Select Interpreter"
  3. Select the interpreter that corresponds to your new environment (the name of the environment will be in parentheses)

Using Python

Scripts

- To create a new Python script (a file with a `.py` extension):
  1. File -> New File -> Python File
- These scripts are generally executed from top to bottom, and are useful for writing longer chunks of code
- You can run a script with the play button in the top right corner of VS Code

Using Python

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Console

- The console is useful for executing smaller bits of code
- To open the console:
    1. Open the Command Palette -> search "Python: Start Terminal REPL"
- Type in a line of code and hit `enter` (or `return` on MacOS) to execute it

Jupyter Notebooks

- **Jupyter notebooks** are a great way to combine code, text, and visualizations in one file
- To create a new Jupyter notebook (a file with a `.ipynb` extension):
  1. File -> New File -> Jupyter Notebook
  2. Set the **kernel** to the Conda environment you created earlier
- Notebooks are composed of **cells** that can be run independently of one another.
  - Cells can either be **code** or **markdown** (text)
- Click the play button beside a given cell to run it

Section 2

The Basics

Variables

- In programming, a variable is a space in computer memory where values may be stored

Defining and Printing Variables

```python
x = 10

print(x)
```

```
10
```

Variables

- Variable names should be short but descriptive (e.g., a variable for the mean of `x` should be named `mean_x`)
  - We are using `x` for pedagogical purposes... In practice, `x` is likely non-descriptive!

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- When a variable name contains multiple words (e.g., apple size), there are a couple of approaches:
    - Use an underscore (e.g., `apple_size`)
    - Use "camel case" (e.g., `appleSize`)
        - Camel case has the first word all lowercase, with subsequent words having their first letter capitalized

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    - Camel case has the first word all lowercase, with subsequent words having their first letter capitalized
- If a variable is not meant to change, it is called a constant. These variables are usually named in all-caps, e.g.:

```
PI = 3.14
```

Variables

Important

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  - E.g., do not name a variable "`print`"
  - We will encounter more names that have been "taken"

Variables

! Important

- Variable names cannot contain spaces (hence the use of camelCase and underscores)
- Variable names cannot start with a number
- Avoid naming variables with names that have already been "taken"
  - E.g., do not name a variable "`print`"
  - We will encounter more names that have been "taken"
- Do not capitalize the first letter of a variable unless it is a `class` (beyond the scope of this workshop)

Variables

Arithmetic

```python
a = 1
b = 2

print(a + b)    # * for multiplication, / for division
```

```
3
```

Note

- Use # to make a **comment**

Variables

Updating Variables

```
x = 10
x = 20
x = 15

print(x)
```

```
15
```

Variable Types

1. `int` (integer; "whole number")
    - less memory
    - 1 is an `int`
2. `float` (real; "decimal number")
    - 1.2 is a `float`
    - 1.0 is a `float`
3. `str` (string; characters/words with no numerical meaning)
    - `"Hello world!"` is a string
    - `"416-767-6767"` is also a string
4. `bool` (boolean; can be `True` or `False`; more on this later)
5. `None`
    - A special type of variable that essentially means "nothing meaningful is stored"
    - Useful as a placeholder

Variable Types

Checking Type

```
x = 416
print(x, type(x))    # type() to check type
```

```
416 <class 'int'>
```

Converting Type

```
x = 416
x = str(x)       # str() to convert to string
print(x, type(x))
```

```
416 <class 'str'>
```

- Can also use `int()` and `float()` to convert to respective type

A Brief Note

- `print()`, `type()`, `str()`, etc. are all examples of **functions**
  - These are essentially pre-made chunks of code for you to use
  - More on this later!

What will the below code output?

```
x = "Variable"

print(type(x))
print(x)
```

Section 3

More Advanced Variable Types

Lists

- Multiple values can be stored in one variable. A list is one way to do this:

Defining Lists

```
x = [0, 1, 2, 3, 4]      # Define a list; square brackets
print(type(x))
```

```
<class 'list'>
```

Lists: Indexing

```
x = [0,1,2,3,4]
```

- To extract one or more values from a list, you **index** the list
- Python uses **zero-based indexing**
  - This means that the first element of a list is index 0, the second element is index 1, ...
  - Element x is index $x - 1$

Lists: Indexing

Indexing Lists

```python
x = [0, 1, 2, 3, 4]
y = 2

print(x[0])      # Index first element of a list
print(x[-1])     # Index last element
print(x[y])      # Index yth element
```

```
0
4
2
```

Lists: Indexing

- You can also take more than one element, or a **slice**, of the list

Slicing Lists

```
x = [0, 1, 2, 3, 4]

print(x[0:2])   # First and second (but not third) elements
```

```
[0, 1]
```

! Important

- `[inclusive:exclusive]`
  - The index on the *left-hand* side of the slice is *included*, whereas the index on the *right-hand* side of the slice is *excluded*

Lists: Indexing

Slicing Lists, continued

```
x = [0, 1, 2, 3, 4]

print(x[0:3]) # First three elements
print(x[0:1]) # Silly way to index just the first element
```

```
[0, 1, 2]
[0]
```

Lists: Length

- To find the length of a list:

len() function

```
x = [0, 1, 2, 3, 4]

print(len(x))   # The length of list x

# Another way to find the last element (not so silly!)
print(x[len(x) - 1])
```

```
5
4
```

Lists: Methods

- Lists have certain functions that can be applied to them, called **methods**
  - Methods are more general than this (e.g., string methods; more on these in a little bit!)

.append()

```python
x = [0,1,2,3,4]

x.append(5)
print(x)
```

```
[0, 1, 2, 3, 4, 5]
```

- Adds an entry to a list

Lists: Methods

.extend()

```
x = [0,1,2,3,4]

x.extend([6,7,8,9])
print(x)
```

```
[0, 1, 2, 3, 4, 6, 7, 8, 9]
```

- Extend a list by “glueing” it to another list

Lists: Methods

Difference Between .append() and .extend()

```
x = [0,1,2,4]
y = [5,6,7,8]

x.append(y)
print(x)

x = [0,1,2,4]    # Redefine x
x.extend(y)
print(x)
```

```
[0, 1, 2, 4, [5, 6, 7, 8]]
[0, 1, 2, 4, 5, 6, 7, 8]
```

Lists: Changing Values

Changing a Value in a List

```
x = [0, 1, 2, 3, 4]

x[0] = 9
print(x)
```

```
[9, 1, 2, 3, 4]
```

Tuples

- Like lists, tuples contain multiple values
- Unlike lists, tuples are **immutable**; they cannot be changed once created.
  - Lists are *mutable*

Defining a Tuple

```
x = ("a", "tuple")  # Round brackets

print(x[0], x[1])
print(type(x))
```

```
a tuple
<class 'tuple'>
```

Tuples

Changing Values... ERROR!

```
x = ("a", "tuple")

x[0] = "Not a"
```

```
TypeError: 'tuple' object does not support item assignment
-----------------------------------------------------------------
TypeError                                    Traceback (most rece
Cell In[19], line 3
      1 x = ("a", "tuple")
      2
----> 3 x[0] = "Not a"
TypeError: 'tuple' object does not support item assignment
```

Tuples

- Tuples have special properties, one of which is below:

Unpacking Tuples

```python
x, y = (6, 7)

print(x)
print(y)
```

```
6
7
```

Section 4

Strings

Strings

- Strings are a lot like lists in that they are **iterable**
  - For now, think of these (lists, strings) as a sequence of values

Strings are Like Lists!

```python
cringe = "Hello world!"

print(cringe[0:5])
print(len(cringe))
```

```
Hello
12
```

Strings

- We can join two strings together through **concatenation**

Concatenation

```python
a = "Hello"
b = "World!"

c = a + " " + b      # Note the space between words
print(c)
```

```
Hello World!
```

Strings

- Sometimes you need to include quotations or other characters inside a string that would normally return an error
- This can be done through an **escape sequence**

ERROR! Why You Need Escape Sequences...

```
x = "This is a "string""
```

```
SyntaxError: invalid syntax (886282512.py, line 1)
  Cell In[23], line 1
    x = "This is a "string""
                   ^
SyntaxError: invalid syntax
```

Strings

Escape Sequences

```python
x = "This is a \"string\""
print(x)
```

```
This is a "string"
```

Note

- The \ indicates the start of an escape sequence

Strings

- Strings are also kind of like tuples in that they are **immutable**

String Immutability

```
cringe = "Hello world!"
cringe[0:5] = "Goodbye"      # Error
```

```
TypeError: 'str' object does not support item assignment
---------------------------------------------------------------------------
TypeError                                 Traceback (most rece
Cell In[25], line 2
      1 cringe = "Hello world!"
----> 2 cringe[0:5] = "Goodbye"      # Error
TypeError: 'str' object does not support item assignment
```

Strings

- We can turn to string **methods** to "modify" strings when needed (they just create a new string "behind the scenes")

.replace()

```
cringe = "Hello world!"

print(cringe.replace("Hello", "Goodbye"))
```

```
Goodbye world!
```

Strings

.split()

```
meme_str = "Harambe, 67, Jar Jar Binks, The Rizzler"

# Split on ", "; returns a list
new_list = meme_str.split(", ")

print(new_list)
print(new_list[0]) # Index the list
```

```
['Harambe', '67', 'Jar Jar Binks', 'The Rizzler']
Harambe
```

How might you index "The Rizzler"?

```
meme_str = "Harambe, 67, Jar Jar Binks, The Rizzler"
new_list = meme_str.split(", ")
```

Strings

How might you index “The Rizzler”?

```python
meme_str = "Harambe, 67, Jar Jar Binks, The Rizzler"
new_list = meme_str.split(", ")
```

```python
print(new_list[3])
print(new_list[-1])
print(new_list[len(new_list) - 1])
```

```
The Rizzler
The Rizzler
The Rizzler
```

Strings

- If you wanted to include variables within text, format strings as per the following:

f-string Example

```python
PI = 3.14159

print(f'Approximate value of pi is {PI}')

print(f'To three decimal places is {PI:.3f}')
```

```
Approximate value of pi is 3.14159
To three decimal places is 3.142
```

Strings

- The string (surrounded by either single `''` or double `""` quotations) is preceded by `f`
- The variable is surrounded by curly braces `{}` within the string
- Formatting is applied to the variable *within the curly braces*
  - A : signals that the proceeding formatting code should be applied to the variable

f-string Formatting Basics

- `PI:.3f`
  - Format variable `PI`
  - Round to 3 decimal places
  - `f` means `float`

Section 5

Booleans

Booleans

- A boolean variable (type `bool`) can be either `True` (1) or `False` (0).

Booleans

```
x = True
y = 10
print(type(x), type(y))

print(y > 5)
print(y < 5)
print(type(y < 5))
```

```
<class 'bool'> <class 'int'>
True
False
<class 'bool'>
```

Booleans

Booleans

```python
x = 5
y = 10

print(x == 5)
print(x > 2 and y > 2)  # Both need to be True
print(x > 2 and y < 2)
print(x > 2 and not y < 2)
print(x > 2 or y < 2)   # At least one needs to be True
```

```
True
True
False
True
True
```

Booleans

Comparison Operators

- > greater than
- < less than
- >= greater than or equal to
- <= less than or equal to
- == equal to
- != not equal to

Logical Operators

- `and`
- `or`
- `not`

Section 6

Conditional Statements

Conditional Statements

- If thing1 happens, do x
- If thing2 happens instead, do y
- Otherwise, do z

Conditional Statements

- If thing1 happens, do x
- If thing2 happens instead, do y
- Otherwise, do z

Basic if/elif/else

```
x = 10  # Value presumably unknown
y = 20

if x == 4:
    print("x is 4")
elif x > y:
    print("x is bigger than y")
else:
    print("none of the above")
```

```
none of the above
```

Conditional Statements

Structure of Conditionals

```python
if x == 4:
    print("x is 4")
```

- Begins with `if`/`elif` ("else if")/`else`
- Boolean expression/value
- Colon :
- Indented code underneath that runs if condition is met

Section 7

Loops

`while` Loops

- Loops are used to run a chunk of code repeatedly
  - Often a set number of times
- This is best shown and unpacked through examples:

while Loops

while Loop Basics

```python
count = 1   # Initialize counter

while count <= 5:
    print(count)
    count = count + 1   # Increment counter
```

1
2
3
4
5

while Loops

Iterating through a list

```
x = [0,1,2,3]
count = 0

while count <= len(x) - 1:
    print(x[count])
    count += 1     # Same as count = count + 1
```

```
0
1
2
3
```

while Loops

- The break statement is used to exit a loop

break

```python
x = 0

while True:      # Infinite loop...
    if x > 10:
        break    # Until you break!
    x += 1

print(x)
```

```
11
```

for Loops

- Often used with the range() function to loop over code a set number of times

for Loops, range()

```
for i in range(3):
    print(i)
```

0
1
2

for Loops

Iterating Over Entries in a List

```
x = [10,20,30,40]

for i in x:
    print(i)
```

```
10
20
30
40
```

for Loops

- Use `enumerate()` and a second indexing variable (usually `i` or `j`) if you want to keep variable position

enumerate()

```
x = [10,20,30,40]

for i, j in enumerate(x):
    print(i, j)
```

```
0 10
1 20
2 30
3 40
```

for Loops

- Loops (and conditionals) may be nested inside one another (like Russian dolls / Matryoshka)

Nesting

```python
x = ["big", "small"]
y = ["cat", "dog"]

for i in x:
    for j in y:
        print(i, j)
```

```
big cat
big dog
small cat
small dog
```